Skill Navora

TRACK 11

Capstone, Portfolio & Interviews

End-to-end capstone, portfolio and GitHub, analyst and data scientist interview preparation with model answers, and a 90-day plan.

10 lessons | Version 2026.10 | Updated October 2026

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Every entry is a clickable link. The PDF also has bookmarks for each lesson and section.

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CAPSTONE, PORTFOLIO & INTERVIEWS · LESSON 1

Portfolio, GitHub and Resume for Data Roles

Turn your learning into evidence employers trust. Choose three to five portfolio projects, structure them on GitHub with strong READMEs, publish dashboards and apps, write an ATS-friendly one-page resume with quantified bullets, and optimise LinkedIn and Naukri profiles for analyst, data scientist, data engineer and GenAI roles.

Career track | ✦ 2026 hiring practice in India | Beginner | ⏱ 90 min

By the end of this lesson you can

- Select portfolio projects that match the target role
- Publish projects on GitHub with professional READMEs and live demos
- Write a one-page, ATS-friendly resume with quantified impact bullets
- Optimise LinkedIn and job-portal profiles and keywords

What hiring managers look for



A portfolio answers three questions for a recruiter in under two minutes.

Target role	Must-have evidence
Data / business analyst	SQL case study, Excel model, Power BI dashboard, EDA with recommendations
Data scientist	EDA, A/B test analysis, ML model with business threshold, explainability
Data engineer	Pipeline (ingest → transform → warehouse), dbt/SQL models, orchestration, tests
GenAI / AI engineer	RAG assistant with evaluation, tool-using agent, deployed demo, cost analysis

Pick 3–5 projects (from this handbook)

1. **SQL:** end-to-end sales case study (SQL lesson 15).
2. **Power BI:** Adventure Cycles dashboard with DAX and drill-through (Power BI capstone).
3. **Python EDA:** telecom churn EDA with recommendations (Python lesson 22) or retail analytics capstone (Python lesson 29).
4. **ML:** churn prediction with profit-based threshold and SHAP (ML lesson 28).
5. **GenAI:** NovaMart RAG support assistant with evaluation and guardrails (GenAI lessons 5, 9, 12).

Make at least one project your own: a dataset from data.gov.in, Kaggle or your workplace (anonymised) about a domain you care about.

GitHub structure and README

```text
churn-prediction/
├── README.md            ← the most important file
├── data/README.md       ← source, licence, how to download (do not commit large/private data)
├── notebooks/01_eda.ipynb
├── src/features.py, train.py, evaluate.py
├── reports/figures/     ← 2–4 key charts used in the README
├── requirements.txt
└── app/ (optional Streamlit demo)
```

README template: **Problem** (one paragraph, business framing) → **Data** → **Approach** (bullets) → **Results** (numbers + one chart) → **Recommendation** → **How to run** → **Next steps**. Put a live link (Streamlit Cloud, Hugging Face Spaces, Power BI publish-to-web with sample data, Tableau Public) at the top.

Resume: one page, quantified

Weak bullet	Strong bullet
"Worked on churn analysis using Python."	"Built a churn model (ROC-AUC 0.80) and targeting rule that reaches 2.3× more churners than random outreach, worth ≈₹60K/month in saved margin."
"Made Power BI dashboards."	"Designed a Power BI sales dashboard (star schema, 25 DAX measures) used weekly by 4 regional managers; cut reporting time from 6 hours to 30 minutes."
"Know SQL."	"Wrote 40+ SQL queries (CTEs, window functions) to analyse 6,300 orders; identified tables/bookcases as loss-making due to >20% discounts."

Formula: **action verb + what you did + tool + measurable result**. Sections: header (name, city, phone, email, LinkedIn, GitHub, portfolio), summary (2 lines, target role), skills (grouped), projects/experience, education, certifications. Use a simple single-column layout (ATS systems mis-read tables and graphics), PDF format, and mirror keywords from the job description honestly.

LinkedIn and job portals

- Headline: "Data Analyst | SQL · Power BI · Python | Turning sales data into decisions".
- About: 3 short paragraphs: who you are, what you build (link projects), what you are looking for.
- Featured: pin your best project and dashboard. Post a short write-up per project (problem, chart, insight).
- Naukri/Instahyre: complete every field, update weekly (recency boosts ranking), set salary and notice period honestly.

KEY TAKEAWAYS

- Choose 3–5 projects that match the target role and show business impact.
- Publish on GitHub with clear READMEs, charts and live demos.
- One-page ATS-friendly resume; every bullet = action + tool + quantified result.
- Keep LinkedIn and portals complete, keyword-aligned and active.

Quick quiz

4 questions · tap an answer to check it

1. What is the most important file in a portfolio repository?

- **A** README.md
- **B** requirements.txt
- **C** .gitignore
- **D** The largest notebook

Answer: A. Recruiters read the README first.

2. Which resume bullet is strongest?

- **A** "Used Python for analysis."
- **B** "Built a churn model (AUC 0.80) that reaches 2.3× more churners than random outreach."
- **C** "Familiar with machine learning."
- **D** "Responsible for dashboards."

Answer: B. Action, tool and quantified result.

3. Why avoid two-column graphic resumes for online applications?

- **A** Applicant tracking systems may parse them incorrectly
- **B** They look unprofessional
- **C** They are too short
- **D** Recruiters dislike colour

Answer: A. Keep layout simple.

4. How many portfolio projects are ideal?

- **A** 3–5 strong, role-relevant projects
- **B** 1
- **C** 20 small ones
- **D** None

Answer: A. Depth beats count.

EXERCISE **Practice**

(1) Choose your target role and list the 4 projects you will publish. (2) Rewrite three of your current resume bullets with the formula. (3) Write the README for one project using the template.

Show solution (worked answer)

A good result: a one-page resume, four public repositories with READMEs and charts, one live demo link and an updated LinkedIn headline.

CAPSTONE, PORTFOLIO & INTERVIEWS · LESSON 2

Data Analyst Interview Preparation: SQL, Excel, Power BI, Statistics and Case Questions

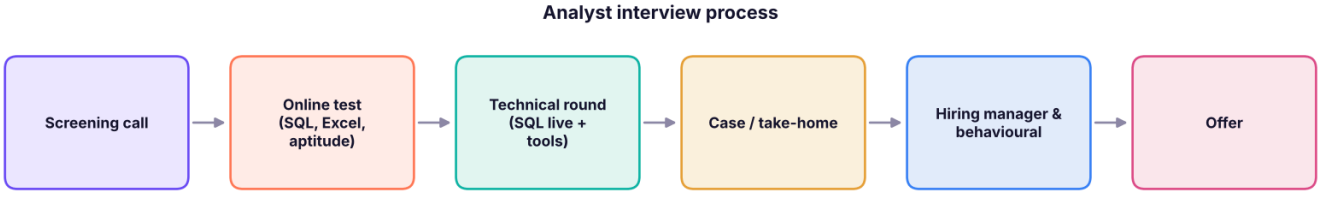
Prepare for analyst interviews with the question types that appear most often, model answers and practice links: SQL coding rounds (joins, aggregation, window functions), Excel and Power BI questions, statistics and A/B testing, metric and product-sense cases, and behavioural questions with the STAR method.

Career track · 2026 · Intermediate · 120 min

By the end of this lesson you can

- Solve common SQL interview patterns quickly and explain your approach
- Answer core Excel, Power BI and DAX questions
- Explain key statistics and A/B testing concepts clearly
- Structure metric, root-cause and product-sense cases
- Tell project stories with STAR

The interview loop



A typical analyst hiring process.

SQL patterns you must know

Python

```
import duckdb, pandas as pd
sales = pd.read_csv("superstore_sales.csv", parse_dates=["order_date"])
print(duckdb.sql("""
-- Top 2 products by sales in each region (window function)
WITH r AS (
  SELECT region, product_name, SUM(sales) AS sales,
         RANK() OVER (PARTITION BY region ORDER BY SUM(sales) DESC) AS rnk
  FROM sales GROUP BY region, product_name)
SELECT region, product_name, ROUND(sales) AS sales FROM r WHERE rnk ⩽ 2 ORDER BY region,
rnk;
""").df())
```

OUTPUT

	region	product_name	sales
0	East	Apex Copier 347	4191180.0
1	East	Zenith Copier 719	3323372.0
2	North	Orbit Copier 746	3790940.0
3	North	Zenith Machine 401	3566453.0
4	South	Orbit Copier 746	5405517.0
5	South	Vertex Copier 262	4258600.0
6	West	Zenith Copier 719	4087068.0
7	West	Zenith Machine 401	4023468.0

Python

```
import duckdb, pandas as pd
sales = pd.read_csv("superstore_sales.csv", parse_dates=["order_date"])
print(duckdb.sql("""
-- Month-over-month growth with LAG
WITH m AS (SELECT DATE_TRUNC('month', order_date) AS month, SUM(sales) AS sales FROM sales
GROUP BY 1)
SELECT month, ROUND(sales) AS sales, ROUND(100.0 * (sales - LAG(sales) OVER (ORDER BY
month)) / LAG(sales) OVER (ORDER BY month), 1) AS mom_pct
FROM m ORDER BY month DESC LIMIT 4;
""").df())
```

OUTPUT

	month	sales	mom_pct
0	2025-12-01	7652267.0	-25.6
1	2025-11-01	10284516.0	67.5
2	2025-10-01	6140290.0	-27.5
3	2025-09-01	8468060.0	45.9

Pattern	Typical question
GROUP BY + HAVING	Customers with more than 5 orders
Joins (inner, left, anti)	Customers who never ordered
Window: ROW_NUMBER/RANK	Top N per group, de-duplicate latest record
Window: LAG/LEAD, running SUM	Growth, retention, cumulative totals
CASE WHEN	Buckets, conditional counts, pivots
Date functions	Monthly cohorts, days between events
Self join / recursive CTE	Manager hierarchy, consecutive days

Approach out loud: clarify the table grain → write the query step by step (CTEs) → check edge cases (NULLs, ties, duplicates) → validate with a small count.

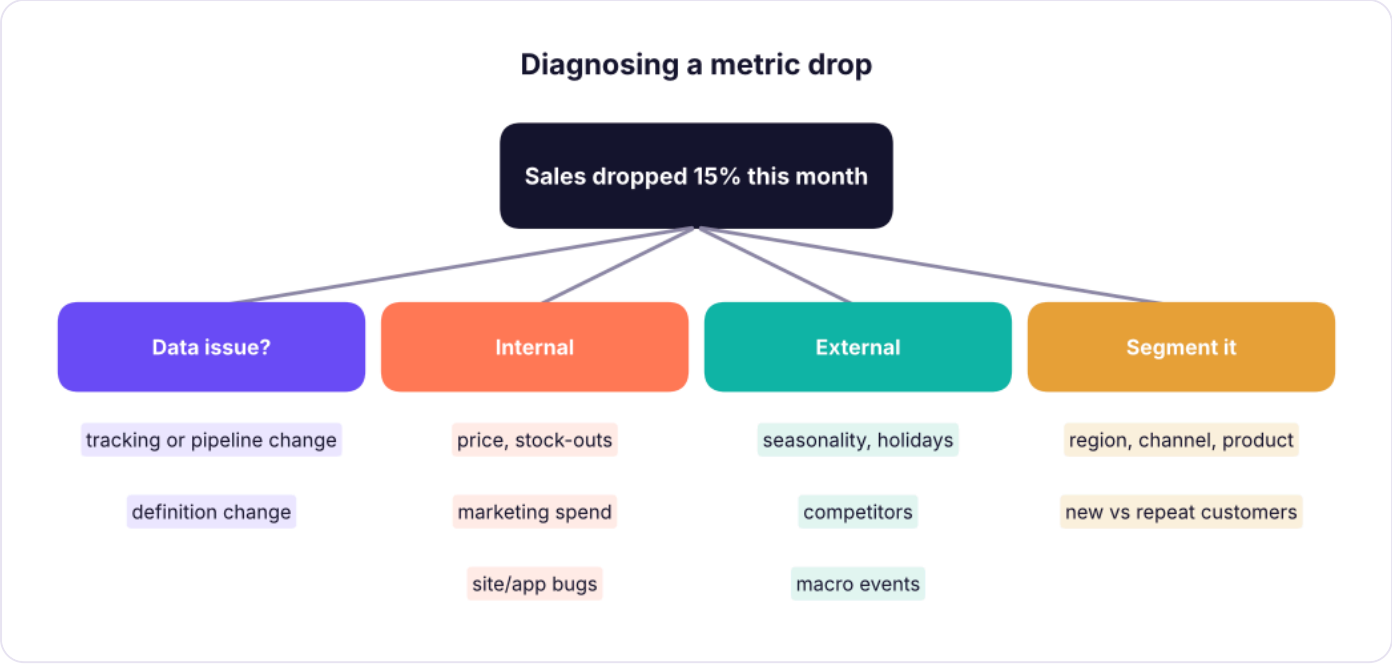
Excel and Power BI questions

Question	Short answer
VLOOKUP vs XLOOKUP vs INDEX-MATCH?	XLOOKUP looks left/right, has defaults and exact match by default; INDEX-MATCH works in older versions; VLOOKUP breaks when columns move
Pivot table vs Power Pivot?	Power Pivot uses a data model with relationships and DAX measures across tables
Calculated column vs measure (DAX)?	Columns are computed row by row and stored; measures are evaluated in filter context at query time
What does CALCULATE do?	Evaluates an expression in a modified filter context
Star schema?	Fact table of events joined to dimension tables; simpler, faster models
Import vs DirectQuery?	Import copies data (fast, scheduled refresh); DirectQuery queries the source live (fresh, slower)
Row-level security?	Roles with DAX filters so users see only their rows

Statistics and experiments

Be ready to explain: mean vs median, standard deviation, distributions, correlation vs causation, confidence intervals, p-values, Type I/II errors, sample size and A/B test pitfalls (peeking, SRM, novelty). See ML lessons 1–6 for model answers.

Case questions: a framework



Root-cause framework for "metric X dropped".

1. Clarify the metric and time frame. 2. Check data quality first. 3. Segment (where is the drop?). 4. Form hypotheses (internal vs external). 5. Validate with data. 6. Recommend action and how you would measure it.

Product-sense metrics: define a north-star metric, input metrics and guardrails (e.g. for a food-delivery app: weekly ordering customers; conversion, delivery time, order value; cancellations, complaints).

Behavioural: STAR

Situation (context) → **Task** (your goal) → **Action** (what *you* did, with tools) → **Result** (numbers, learning). Prepare 6 stories: a project you are proud of, a mistake you fixed, a conflict, a tight deadline, influencing without authority, learning a new tool fast.

Practice resources

W3Schools SQL quiz and exercises, LeetCode SQL 50, HackerRank SQL, StrataScratch, DataLemur, Kaggle Learn; mock interviews with peers; time yourself (15 minutes per medium SQL question).

KEY TAKEAWAYS

- Master SQL patterns: aggregation, joins, window functions, CTEs, dates; think aloud and check edge cases.
- Know Excel lookups/pivots and Power BI modelling, DAX, CALCULATE, star schema and RLS.
- Explain statistics and A/B testing simply.
- Use a structured framework for metric and product cases.
- Prepare STAR stories with quantified results.

Quick quiz

4 questions · tap an answer to check it

1. Which SQL feature finds the top 3 products in each category?

A A window function such as RANK() OVER (PARTITION BY category ORDER BY sales DESC)

B GROUP BY only

C DISTINCT

D UNION

Answer: A. Then filter rank ≤ 3.

2. Sales dropped 15%. What is the first check?

A Whether the data or tracking changed

B Fire the marketing team

C Build an ML model

D Ignore it

Answer: A. Rule out data issues first.

3. What is a DAX measure?

- **A** An expression evaluated at query time in the current filter context
- **B** A stored column
- **C** A table relationship
- **D** A Power Query step

Answer: A. Columns are stored row by row.

4. What does STAR stand for?

- **A** Situation, Task, Action, Result
- **B** Skills, Tools, Analysis, Reporting
- **C** Start, Test, Assess, Repeat
- **D** None

Answer: A. Structure for behavioural answers.

EXERCISE **Practice**

(1) Write SQL for: customers who ordered in 2024 but not in 2025; the second-highest order per customer; 7-day moving average of sales. (2) Prepare two STAR stories. (3) Do a 30-minute mock case: "Delivery complaints doubled last week."

Show solution (worked answer)

Use NOT EXISTS / LEFT JOIN ... IS NULL for the anti-join, DENSE_RANK() = 2 for the second highest, and AVG(...) OVER (ORDER BY day ROWS BETWEEN 6 PRECEDING AND CURRENT ROW) for the moving average.

CAPSTONE, PORTFOLIO & INTERVIEWS · LESSON 3

Data Scientist, ML and GenAI Interview Preparation: Questions and Model Answers

A compact question bank with model answers for data scientist, ML engineer and GenAI roles: Python and pandas, statistics, ML fundamentals, model evaluation, feature engineering, system and case design (churn, fraud, recommendations), MLOps, LLMs, RAG and agents, plus how to handle take-home assignments and live coding.

Career track · ✦ 2026 · Advanced · ⏱ 120 min

By the end of this lesson you can

- Answer core Python, statistics and ML questions concisely
- Walk through an end-to-end ML system design
- Explain LLM, RAG and agent concepts and trade-offs
- Plan and present take-home assignments

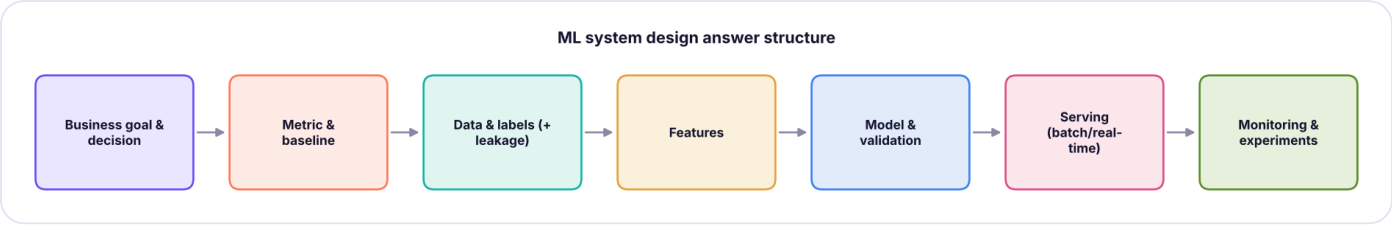
Python and pandas (quick answers)

Question	Answer
List vs tuple vs set vs dict?	Mutable sequence / immutable record / unique unordered / key→value; sets and dicts give O(1) lookups
`apply` vs vectorisation?	Vectorised column operations run in C and are much faster; avoid `apply(axis=1)` on large data
`merge` vs `join` vs `concat`?	merge on columns, join on index, concat stacks
How to handle missing values?	Understand why, then drop, impute (median, group-wise, model) or flag; impute inside CV
Generator vs list?	Generators are lazy and memory-efficient, consumed once

Statistics and ML fundamentals

Question	Model answer
Bias-variance trade-off?	Simple models underfit (bias), complex models overfit (variance); choose complexity by validation error
L1 vs L2?	L1 zeroes coefficients (selection); L2 shrinks smoothly; Elastic Net mixes
Why cross-validation?	A single split is noisy; k-fold averages and supports tuning without touching the test set
Precision vs recall?	Precision: flagged that are right; recall: positives caught; choose by error costs
ROC-AUC vs PR-AUC?	PR-AUC is more informative for rare positives
Bagging vs boosting?	Parallel averaging (variance) vs sequential error correction (bias)
How does gradient descent work?	Iteratively step weights against the loss gradient with a learning rate
What is data leakage?	Using information unavailable at prediction time; prevent with pipelines, time-based splits and feature cut-offs
p-value?	Probability of data at least this extreme if H0 were true; not the probability H0 is true

ML system design: a template



Answer ML design questions in this order.

Example: **"Design a fraud detection system for UPI payments."** Real-time decision under 100 ms; label = confirmed fraud (delayed, imbalanced); features: amount vs user history, device/location change, merchant risk, velocity, graph links; model: gradient boosting + rules + anomaly scores; evaluate PR-AUC and recall at alert capacity; serve with a feature store and low-latency API; monitor drift and attack patterns; human review queue; A/B test thresholds.

LLM, RAG and agent questions

Question	Model answer
How do LLMs work?	Transformers trained to predict the next token, then instruction-tuned and aligned with feedback
RAG vs fine-tuning?	RAG for changing knowledge with citations; fine-tuning for behaviour/format
How to evaluate a RAG system?	Retrieval hit rate/MRR + answer correctness, faithfulness, abstention on a labelled set
How to reduce hallucinations?	Grounding, tools, "say I don't know", citations, validation, evaluation
What is an agent?	An LLM in a loop choosing tool actions toward a goal, with memory and stopping conditions
What is MCP?	Open protocol to connect assistants to tools/data via servers (tools, resources, prompts)
Prompt injection defence?	Separate data from instructions, least-privilege tools, approvals, output checks
Cost control?	Model routing, caching, batching, shorter context and outputs

Take-home assignments

1. Restate the problem and assumptions in the first lines of your report.
2. Do a focused EDA; do not chart everything.
3. Baseline first, then 1–2 better models; validate correctly.
4. Translate results into business impact and a recommendation.
5. Clean, runnable code (requirements, README); a 5-slide summary.
6. State limitations and next steps honestly.

Live coding tips

Clarify inputs/outputs and edge cases, start with a simple correct solution, test with a small example, then optimise and state Big-O (Python DSA lessons 26–28).

KEY TAKEAWAYS

- Answer concisely, then go deeper only if asked.
- Use the design template: goal → metric → data → features → model → serving → monitoring.
- Know LLM/RAG/agent trade-offs, evaluation and safety.
- Take-homes: baseline, correct validation, business recommendation, clean code, honest limitations.

Quick quiz

4 questions · tap an answer to check it

1. What should you state first in an ML system-design answer?

A The business goal and the decision the model supports

B The algorithm

C The GPU type

D The library version

Answer: A. Framing first.

2. Best metric for a fraud model with 0.5% fraud?

A PR-AUC and recall at the alert capacity

B Accuracy

C R^2

D MAE

Answer: A. Rare positives.

3. RAG or fine-tuning for policy documents that change monthly?

- **A** RAG
- **B** Fine-tuning
- **C** Neither
- **D** Training from scratch

Answer: A. Knowledge belongs in retrieval.

4. What must a take-home report include besides the model?

- **A** Business impact, recommendation and limitations
- **B** Only code
- **C** Only accuracy
- **D** Every chart possible

Answer: A. Decision-makers read conclusions.

EXERCISE **Practice**

(1) Answer "Design a product recommendation system for an e-commerce home page" using the template. (2) Explain RAG to a non-technical manager in 60 seconds. (3) Do one timed take-home on the telecom churn data (3 hours).

Show solution (worked answer)

Recommendations: goal = revenue per session; candidates via collaborative filtering/embeddings + popularity; ranking model with user, item and context features; offline metrics (recall@k, NDCG) + online A/B test on revenue; cold start with popular/contextual items; monitor diversity and drift.

CAPSTONE, PORTFOLIO & INTERVIEWS · LESSON 4

Your 90-Day Plan: Learning Roadmap, Job Search Strategy and Growing in Your First Role

Turn this handbook into a job. A week-by-week 90-day study and project plan for analyst, data science and GenAI tracks, a weekly routine, a job-search funnel with referrals and outreach messages, salary research and negotiation basics, and habits that help you grow in the first 6 months on the job.

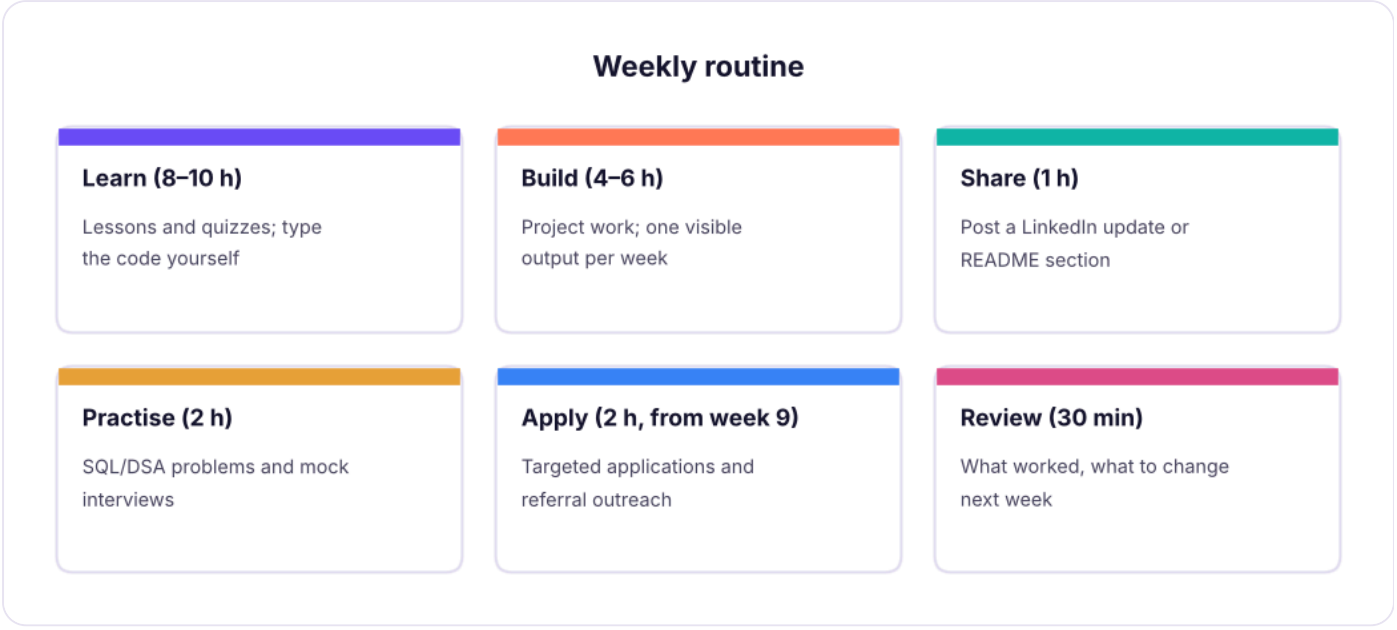
Career track · 2026 · Beginner · 60 min

By the end of this lesson you can

- Follow a realistic 90-day roadmap with weekly goals
- Run a job-search funnel with referrals and targeted applications
- Prepare for salary discussions with market data
- Build habits for success in the first months of a data role

90-day roadmap

Weeks	Analyst track	Data science / ML track	GenAI track
1–2	Excel lessons 1–6	Python 1–10	Python 1–12
3–4	Excel 7–12 + SQL 1–5	Python 15–22 (NumPy, pandas, EDA)	Python 15–19, 23–24 (APIs)
5–6	SQL 6–15 + SQL case study	Stats ML 1–6 (statistics, A/B)	GenAI 1–4 (LLMs, prompts, APIs, embeddings)
7–8	Power BI 1–6 + dashboard project	ML 7–15 (workflow, regression, metrics, CV)	GenAI 5–8 (RAG, tools, agents, MCP)
9–10	Python 1–16, 20–22	ML 16–25 + one capstone	GenAI 9–12 + capstone app
11–12	AI analytics 1–4, portfolio polish	ML 26–28, portfolio polish	Portfolio, deployment, write-ups
13	Interview prep (Career 2) and applications	Interview prep (Career 3)	Interview prep (Career 3)



A sustainable weekly routine.

Job-search funnel

Stage	Target per week	Tips
Research companies and roles	15	Match your projects to their domain
Referral / recruiter outreach	10	Personalised messages; alumni and community contacts
Applications	10–15	Tailor resume keywords; apply within 3 days of posting
Screens and tests	2–4	Track results; review mistakes
Interviews	1–3	Prepare company-specific cases

Referral message template:

```text
Hi <Name>, I'm a data analyst (SQL, Power BI, Python) who built <project link> on <domain>.
I saw the <role> opening at <company> (job id). Could you refer me or share what the team
values most?
Happy to send my one-page resume. Thank you!
```

Track everything in a simple sheet: company, role, date, source, status, next step, notes.

Salary and offers

Research ranges on AmbitionBox, Glassdoor, LinkedIn Salary and peers for your city and experience. Ask for the full CTC breakdown (fixed, variable, joining bonus, benefits), negotiate on evidence (skills, competing offers, market data), and compare learning opportunities, team quality and growth, not only salary.

First 90 days in the job

1. Learn the business: products, customers, key metrics and their definitions.
2. Find the data: sources, owners, known quality issues; document as you go.
3. Deliver a small, useful win in the first month.
4. Communicate: weekly updates, clear charts with insight titles, assumptions stated.
5. Automate repetitive reports; share reusable SQL and notebooks.
6. Keep learning: one new skill per quarter (dbt, cloud, experimentation, GenAI).

KEY TAKEAWAYS

- Follow the 90-day roadmap with a weekly routine of learn, build, share, practise and apply.
- Run the job search as a funnel with referrals, tailored applications and tracking.
- Negotiate with market data and evaluate the whole offer.
- In the new role, learn the business, deliver early wins, communicate clearly and keep learning.

Quick quiz

3 questions · tap an answer to check it

1. What should each week of the plan produce?

A At least one visible output (project progress, write-up or dashboard)

B Only reading

C Only certificates

D Nothing until week 12

Answer: A. Evidence compounds.

2. Which outreach is most effective?

A Personalised referral requests linking a relevant project

B Generic mass messages

C Applying to every job without reading it

D Waiting for recruiters

Answer: A. Relevance wins.

3. What matters when comparing offers?

- **A** Full CTC breakdown plus learning, team and growth
- **B** Only the fixed salary
- **C** Only the company logo
- **D** Only the job title

Answer: A. Look at the whole package.

EXERCISE **Practice**

Write your personal 13-week plan with weekly goals, choose your capstone, create your job-tracker sheet, and send three referral messages this week.

Show solution (worked answer)

A good plan names the lessons, the project deliverable and the job-search target for every week, with time blocked in your calendar.

CAPSTONE, PORTFOLIO & INTERVIEWS · LESSON 5

Data Engineering and Analytics Interview Questions with Model Answers

Practise twelve core questions on ETL versus ELT, idempotency, batch and streaming, schemas, surrogate keys, late data, dbt tests, Airflow, Spark shuffles, warehouses versus lakehouses, pipeline testing and change data capture, each with a model answer and the reasoning an interviewer listens for.

Gap-fill lesson · ✦ 2026 · Intermediate · ⏱ 120 min

By the end of this lesson you can

- Answer core data engineering and analytics questions clearly and briefly
- Give the trade-off behind each answer, not only the definition
- Recognise follow-up questions that test depth

How interviewers use these questions

Interviewers rarely want a textbook definition. They listen for four things: whether you know the term, whether you can explain it simply, whether you know when it applies, and whether you know its trade-off. A good answer covers all four in under two minutes. The model answers below are written at that length.

The twelve questions

1. What is the difference between ETL and ELT? *Model answer.* ETL transforms data before loading it into the target; ELT loads raw data first and transforms it inside the warehouse. ELT suits cloud warehouses that have cheap, powerful compute, and it keeps raw data so models can be rebuilt. ETL is chosen when data must be masked or validated before it lands, for example personal data that must never reach the warehouse in raw form.

2. What is idempotency? *Model answer.* An operation is idempotent when running it twice has the same effect as running it once. Pipelines need this because retries and reruns are normal. A load

that appends rows is not idempotent; a load that merges on a key, or overwrites a partition, is.

3. Batch versus streaming? *Model answer.* Batch processes accumulated data at intervals, such as nightly. Streaming processes each event as it arrives. Batch is simpler, cheaper and easier to test; streaming gives low latency at higher operational cost. Choose by how fresh the data must be and what it costs to be late.

4. Star versus snowflake schema? *Model answer.* Both have a central fact table and dimension tables. A star schema keeps each dimension flat, which makes queries simple and fast. A snowflake normalises dimensions into sub-tables, which saves a little storage and adds joins. Star schemas are usually preferred for reporting; snowflakes appear where dimensions are very large or shared.

5. What is a surrogate key and why use it? *Model answer.* A surrogate key is a system-generated identifier, such as an integer, that has no business meaning. It keeps the warehouse stable when source keys change, reduces join size, and is essential for slowly changing dimensions where one customer has several versions over time.

6. How do you handle late-arriving data? *Model answer.* First, decide the tolerance: how late can a record arrive and still count? Then reprocess the affected partitions or window rather than only appending new rows, use the event time rather than arrival time for aggregates, and report the restated numbers with a note. Monitor how often late data arrives so the window can be tuned.

7. What does a dbt test check? *Model answer.* dbt tests check assumptions about a model's data: that a column is unique, not null, that values belong to an accepted list, or that a relationship between tables holds. They run on every build, and a failing test stops the pipeline from publishing wrong data. Custom tests can check business rules, such as "revenue is never negative".

8. What is Airflow used for? *Model answer.* Airflow schedules and orchestrates workflows defined as directed acyclic graphs (DAGs). It runs tasks in dependency order, retries failures, records run history and alerts on errors. It does not process the data itself; it calls the tools that do.

9. Spark shuffles and partitions? *Model answer.* A shuffle is the redistribution of data across the cluster, typically for a join or a group-by, and it is expensive because data moves over the network. Partitions are the units of parallel work. Too few partitions leave machines idle; too many add overhead. Reducing shuffles, filtering early and choosing sensible partition counts are the usual fixes for slow jobs.

10. Warehouse versus lakehouse? *Model answer.* A warehouse stores structured tables with strong SQL performance and governance. A lakehouse keeps data in open file formats on cheap storage and adds table features such as transactions and schema management, so it can serve both analytics and machine learning. Many organisations use both: a lakehouse for raw and mixed data, and a warehouse or serving layer for reporting.

11. How do you test a data pipeline? *Model answer.* Test at four levels: unit tests for transformation functions on small inputs; schema tests on each table; data-quality tests such as row counts, nulls

and reconciliation totals against the source; and an end-to-end run on sample data. Add a failure drill, where you break an input on purpose and confirm the alert fires.

12. What is CDC? *Model answer.* Change data capture reads a database's change log to capture inserts, updates and deletes as they happen, instead of re-reading whole tables. It keeps the warehouse close to real time at low load on the source. It needs database access for the log and careful handling of schema changes.

Follow-up questions to expect

If you say...	Expect a follow-up like...
"I would use an upsert"	"What is the key, and what happens with duplicates in the source?"
"Streaming for everything"	"What does it cost to run, and who is paged at night?"
"dbt tests cover quality"	"Which failures would a dbt test miss?"
"Partition the data"	"What partition key would you choose, and why not the most detailed one?"

INTERVIEW CORNER

A reliable way to answer

Give the definition in one sentence, the reason it matters in one sentence, one example, and the trade-off. Then stop and let the interviewer ask for more. A short answer with a clear trade-off impresses more than a long lecture.

::::key What to remember - Every answer needs a trade-off, not only a definition. - Idempotency, late data and testing are the topics that separate candidates. - Use a concrete example from your own work when you can. - Expect follow-ups; have the next layer of detail ready. ::::

Quick quiz

5 questions · tap an answer to check it

1. Which approach is idempotent?

A Appending the same file every night

B Merging rows on a key so a rerun does not duplicate them

C Inserting without checks

D Sending every event twice

Answer: B. Merges on a key keep a rerun from creating duplicates.

2. When is ETL the better choice than ELT?

A When the warehouse is very cheap

B When data must be masked or validated before it reaches the warehouse

C When there is no transformation

D When data is never stored

Answer: B. ETL transforms before loading, which suits strict masking requirements.

3. What is a surrogate key?

A The customer's email address

B A system-generated identifier with no business meaning

C A password

D The date of the record

Answer: B. Surrogate keys keep the warehouse stable when source keys change.

4. What does a dbt `unique` test check?

- **A** That the model runs quickly
- **B** That a column has no duplicate values
- **C** That the file is in CSV format
- **D** That the model is documented

Answer: B. The test fails when duplicates appear in the column.

5. Which tool is responsible for scheduling and retrying the steps of a pipeline?

- **A** Spark
- **B** Airflow
- **C** A dashboard
- **D** A spreadsheet

Answer: B. Airflow orchestrates tasks; the processing tools do the data work.

EXERCISE **Practise aloud**

Choose five of the twelve questions and answer each aloud in under two minutes, with one example from a project you have done. Record yourself, listen back, and rewrite any answer that has no trade-off in it.

Show solution (worked answer)

Hints: time each answer. Replace vague words such as "stuff" or "some tools" with a named tool or a number. If you cannot give an example, use a realistic example from a public dataset and say so.

CAPSTONE, PORTFOLIO & INTERVIEWS · LESSON 6

Business Scenario Interviews: Seven Problems and Answer Frameworks

Practise seven open-ended business scenarios, from a sales decline and rising churn to a slow dashboard, a failed pipeline, an incorrect AI assistant and a vague stakeholder request. Each has a structured answer framework: clarify, hypothesise, analyse, recommend and measure.

Gap-fill lesson · 2026 · Intermediate · 120 min

By the end of this lesson you can

- Structure an open-ended business answer with a repeatable framework
- Ask the clarifying questions that show judgement before analysing
- Propose tests and measures, not only conclusions
- Recognise when the right answer is "I need to clarify the requirement first"

What a scenario interview tests

Scenario questions have no single right answer. The interviewer is judging how you think: whether you clarify before you analyse, whether your hypotheses are specific enough to test, whether you choose the cheapest test that would change a decision, and whether you can recommend an action and say how you will know it worked.

The framework

Use the same five steps for every scenario, and say the step names aloud.

1. **Clarify.** What exactly happened, over what period, for which segment, compared with what? Ask two or three questions before answering.
2. **Hypothesise.** List three to five possible causes, ranked by how likely and how cheap they are to check.
3. **Analyse.** For each hypothesis, name the data that would confirm or reject it, and the test or query you would run.

4. **Recommend.** State the most likely cause and the action, with the trade-off.
5. **Measure.** Name the metric that will show the action worked, and the timeframe.

Scenario 1: sales have declined by 20%

- **Clarify:** Which products, regions and channels? Compared with last month, last year, or the plan? Did prices, stock or the sales team change?
- **Hypotheses:** a data or reporting change; a seasonal effect; lost customers; lower traffic; a stock-out; a price change; a competitor move.
- **Analysis:** split the decline by region, product, channel and customer type; compare new and repeat customers; check whether the drop comes from fewer orders or smaller orders; reconcile the total with the finance ledger.
- **Recommend:** act on the largest confirmed driver only. If the data is wrong, the first action is to fix the data.
- **Measure:** weekly sales in the affected segment against a comparison group.

Scenario 2: customer churn has increased

- **Clarify:** How is churn defined (cancellation, or no order for 90 days)? Which customer groups? Since when?
- **Hypotheses:** a price increase; a product issue; a support problem; a competitor offer; a cohort acquired through a weak channel.
- **Analysis:** cohort retention by sign-up month and channel; the tenure at which customers leave; support tickets before churn; the share of churners who used a discount.
- **Recommend:** a targeted retention offer for the group most at risk, with a control group to measure it.
- **Measure:** retention of the treated group against the control after 60 to 90 days.

Scenario 3: a dashboard is slow

- **Clarify:** How slow, for whom, and since when? Which filters? Is it the first load or each interaction?
- **Hypotheses:** too much data in the model; inefficient measures; too many visuals; a slow data source; unlimited filters.
- **Analysis:** check the query and refresh timings; count the rows loaded; find the measures that scan the most data.
- **Recommend:** aggregate the fact table to the level the dashboard needs, simplify the most expensive measures, and reduce the visuals on the first page.

- **Measure:** time to first render, p95, and user feedback.

Scenario 4: a data pipeline has failed

- **Clarify:** Which job, which step, what error, since when, and what depends on it? Is it the first failure or a repeat?
- **Hypotheses:** a source schema change; a late file; a permission change; a memory limit; a bad record; a changed API.
- **Analysis:** read the logs at the failing step; compare row counts with the previous run; check upstream jobs and the source schema.
- **Recommend:** restore service first (rerun the failed partition if the job is idempotent), then fix the cause and add a test that would have caught it.
- **Measure:** time to detect and time to recover, and whether the same failure recurs.

Scenario 5: an AI assistant gives incorrect answers

- **Clarify:** Which questions, how often, and how wrong? Did the assistant have the right documents? Was the answer checked by a person?
- **Hypotheses:** retrieval missed the right document; the document is out of date; the prompt allows guessing; the model is wrong on a specific topic; a tool returned bad data.
- **Analysis:** build a test set of the failing questions; check whether the right passage was retrieved; read the answers against the sources; check tool outputs.
- **Recommend:** fix the cause that the test points to (retrieval, content, prompt or tool), and add the cases to the regression suite.
- **Measure:** accuracy on the test set and the rate of wrong answers reported by users.

Scenario 6: an enterprise customer wants to automate a manual workflow

- **Clarify:** What is the workflow today, how long does it take, how often, what goes wrong, and who approves each step?
- **Hypotheses:** the workflow is fixed (a workflow tool suffices); it needs judgement at some steps (an agent with approval); the data is not ready.
- **Design:** map the steps; mark each as deterministic, model-driven or needing a person; choose the simplest tool; define the approval points, the data access and the logs.
- **Recommend:** a pilot on one part of the workflow with a clear success measure and a stop rule.
- **Measure:** time per case, error rate, the share handled without a person, and cost per case.

Scenario 7: a stakeholder asks for a dashboard but cannot define the requirements

- **Clarify:** What decision will this dashboard support? Who will use it, and how often? What would they do differently if it showed X?
- **Hypotheses:** the stakeholder knows the problem but not the measures; they need an example before they can specify.
- **Approach:** run a short discovery: interview two users, sketch three layouts on paper, and agree the first three questions the dashboard must answer.
- **Recommend:** build a first version with the three agreed questions, show it, and collect changes.
- **Measure:** whether the stakeholder uses it in a real meeting within a month.

INTERVIEW CORNER

The strongest answer to a vague request

Say that you would not build anything yet. Explain the decision the work must support, propose a 30-minute discovery session, and describe the smallest version you would show within a week. Interviewers reward candidates who protect the team from building the wrong thing.

::::key What to remember - Clarify before analysing; two good questions show judgement. - Rank hypotheses by likelihood and by how cheap they are to test. - Recommend one action with its trade-off, and name the measure that proves it worked. - For a vague request, the first deliverable is a decision and a shared definition. ::::

Quick quiz

5 questions · tap an answer to check it

1. What is the first step in answering a scenario question?

A Recommend an action

B Build a chart

C Clarify what exactly happened, over what period, and compared with what

D Name the tool to use

Answer: C. Analysis without a clear question wastes effort.

2. Which hypothesis should you test first in a sales decline?

A The one you like best

B The likely cause that is cheapest to check, such as a data change

C The one that needs the most budget

D The competitor's product launch

Answer: B. Ranking by likelihood and cost focuses effort.

3. The dashboard is slow. Which fix comes first?

A Buy a faster computer

B Check where time is spent, then aggregate or simplify the costliest parts

C Add more visuals

D Remove all measures

Answer: B. Measure before optimising.

4. A pipeline fails. What is the right order of actions?

A Fix the code and then tell someone

B Restore service where safe, then fix the cause and add a test

C Delete the logs

D Wait for the next scheduled run

Answer: B. Recovery first, then prevention.

5. A stakeholder cannot define a requirement. What should you do?

- **A** Build the most complete dashboard possible
- **B** Refuse the work
- **C** Run a short discovery to agree the decision and the first questions
- **D** Ask for a bigger budget

Answer: C. The first deliverable is a shared definition of the problem.

EXERCISE

Answer two scenarios

Choose two of the seven scenarios and answer each in four minutes using the five-step framework, out loud or in writing. For each, state the first two clarifying questions, your top three hypotheses, the test you would run first, and the measure you would report.

Show solution (worked answer)

Hints: keep the answer within the five headings. Make the first test specific enough to run this week. If you do not know a number, say how you would obtain it.

CAPSTONE, PORTFOLIO & INTERVIEWS · LESSON 7

Behavioural Interviews and STAR Answers

Prepare answers to seven common behavioural questions using the STAR method (Situation, Task, Action, Result). Learn to structure a two-minute story, show your own contribution, include a measurable result, and tell an honest story about a failure or a conflict.

Gap-fill lesson | 2026 | Beginner | 90 min

By the end of this lesson you can

- Structure a behavioural answer using STAR
- Describe your own actions clearly, without taking credit for the team's work
- Include a result, with a number where possible
- Talk honestly about a failed project and what changed afterwards

The STAR method

Behavioural questions ask "tell me about a time when..." The interviewer wants evidence of how you actually work. The **STAR** structure gives a clear story in about two minutes:

Part	Length	What to say
Situation	1–2 sentences	Where you were, what the context was
Task	1 sentence	What you were responsible for
Action	60–90 seconds	What you did, step by step, using "I" for your own work
Result	20–30 seconds	What changed, with a measure, and what you learned

Most weak answers spend too long on the situation and not enough on the action and result. Aim for the action to be the longest part.

CASE STUDY **The same project, told two ways**

A weak answer: "We had a messy spreadsheet and the team was struggling, so we worked on it and it got better." A STAR answer: "Our monthly churn report was built by hand and took two days, with errors found by finance each time. My task was to automate it without changing the numbers. I wrote a cleaning script with checks for each source, compared its output with the manual report for three months, and set up a scheduled run. The report now takes ten minutes and the variance to finance is under 0.5%." The second answer names the action and the measure.

Question 1: tell me about yourself

Use a 90-second version: your current role or study, two relevant skills with an example each, and why this role. End with a sentence connecting to the job. Do not recite your CV.

Question 2: explain a difficult project

Choose a project with real obstacles. Describe the constraint (data quality, a deadline, a changing requirement), the step you took to get past it, and what you would do differently. A project that was hard but went well is a good choice.

Question 3: describe a problem you solved

Pick a problem with a clear cause and a verifiable fix. Show how you diagnosed it: what you checked first, what the evidence showed, and why you chose the fix you did. Quantify the result where you can.

Question 4: explain a conflict with a stakeholder

Describe the disagreement fairly, including the other person's valid concern. Show how you found common ground: a shared definition, a small pilot or a data check. Avoid describing the other person as the problem; describe what you changed in how you worked together.

Question 5: describe a failed project

Be honest. Choose a real failure with a lesson. Explain what you expected, what went wrong, what you did when you noticed, and what you do differently now. Interviewers trust candidates who can

name a mistake and show a changed habit.

> **WATCH OUT** **Do not invent results**
>
> Never state a number you cannot support. If the result is qualitative, say so: "the team adopted it, and two colleagues now use the same check." If you do not know the impact, say how you would measure it. Interviewers often ask a follow-up about the number, and an invented figure falls apart.

Question 6: how do you prioritise work?

Describe a method you actually use: for example, ranking by business impact and effort, asking the requester what decision the work supports, and reviewing the list weekly. Give one example where you said no or postponed a request, and why.

Question 7: how do you explain technical concepts to non-technical stakeholders?

Show a technique: start with the decision, use one concrete example, avoid jargon or define it once, and check understanding by asking the person to restate the implication. Give an example where a chart or a single number replaced a long explanation.

Preparing your stories

Prepare five stories that each cover different themes (a success, a failure, a conflict, a difficult deadline, a learning moment). Each story can answer several questions. Write each in STAR form, time it, and practise until it fits in about two minutes without notes.

Story	Useful for questions
Automated the monthly report	2, 3, 6
Disagreement over a metric definition	4, 7
A pipeline failure that taught a lesson	2, 5
A dashboard that went unused	5, 6, 7
A course or project you completed on your own	1, 3

::::tip Practise with a recording Record your answers on a phone. Check whether the action is the longest part, whether you say "I" for your own work, and whether the result has a number or a clear

change. Re-record until each answer lands in two minutes. ::::

::::key What to remember - Use STAR: short situation and task, a detailed action, and a result with a measure. - Say "I" for what you did; say "we" for the team's work, and be clear about your share. - Never invent numbers; say how you would measure an unknown impact. - Prepare five stories that cover several questions each. ::::

Quick quiz

5 questions · tap an answer to check it

1. Which part of STAR should be the longest?

- **A** Situation
- **B** Task
- **C** Action
- **D** Result

Answer: C. The action is where the interviewer learns how you work.

2. What makes a result strong in a STAR answer?

- **A** It is vague and modest
- **B** It names a change, ideally with a measure
- **C** It describes the team's pride
- **D** It is left out

Answer: B. A measure or clear change shows the impact of your work.

3. **In a question about a failed project, what do interviewers value most?**

- **A** A story with no failure
- **B** A real failure, what you did about it, and a changed habit
- **C** Blaming others
- **D** A list of tools

Answer: B. Honesty and learning are what they listen for.

4. **Why prepare five stories rather than many?**

- **A** Interviews allow only five
- **B** Each story can answer several questions, and five are easier to practise well
- **C** Five is the legal minimum
- **D** Stories should never be reused

Answer: B. A small set of well-practised stories covers most questions.

5. **What should you do if you do not know the impact of a project?**

- **A** Estimate a number that sounds plausible
- **B** Say how you would measure it or describe the qualitative change honestly
- **C** Avoid mentioning the project
- **D** Change the subject

Answer: B. Honest answers survive follow-up questions; invented ones do not.

EXERCISE **Write five stories**

Write five STAR stories from your own experience, study or volunteering: a success, a failure, a conflict, a deadline and a learning moment. Time each one out loud, and map each to the seven questions above.

Show solution (worked answer)

Hints: if you lack workplace examples, use a project you did alone, a course assignment, or a volunteer activity; the structure is the same. Mark the action sentences and check that each contains "I" and a concrete verb.

CAPSTONE, PORTFOLIO & INTERVIEWS · LESSON 8

Eight Capstone Project Briefs and the Portfolio Checklist

Choose one of eight portfolio-level capstone projects covering dashboards, churn prediction, an end-to-end data pipeline, an AI data analyst, an enterprise RAG assistant, an agent for business operations, an MCP business assistant and an industry-specific project. Each brief has requirements, deliverables and an evaluation rubric, and a checklist makes every project presentable to an employer.

Gap-fill lesson · ✦ 2026 · Advanced · ⏱ 150 min

By the end of this lesson you can

- Scope a portfolio project with clear requirements and deliverables
- Choose a project that matches your target role
- Prepare a project so that an employer can review it in ten minutes
- Evaluate your own work against the rubric before you show it

How to use these briefs

Choose **two** projects, not eight: one that matches your target role closely, and one that shows a different skill. Each brief has the same parts: the problem, requirements, deliverables, the data to use, and the rubric the work will be judged against. Read the rubric first; it tells you what matters most.

Project 1: Business analytics dashboard

- **Problem:** a regional sales manager needs to see revenue, margin and returns by region, category and month, and to spot where profit is falling.
- **Requirements:** a clean star schema; at least 8 measures in one measure table; a page for trends, a page for drill-down, and a page of exceptions; a filter for date and region.
- **Deliverables:** the report file, a 1-page guide for users, and a README explaining the data source and refresh.

- **Data:** `superstore_sales.csv`.
- **Rubric:** correctness of totals against a control (30%); clarity of the first page (25%); measures documented and tested (25%); usability on a phone or small screen (20%).

Project 2: Customer churn prediction

- **Problem:** a telecom company wants to know which customers are likely to leave in the next month, so that a retention offer can be targeted.
- **Requirements:** a defined churn label; a train and test split that respects time; a baseline model; one stronger model; evaluation with precision, recall and a business cost; an explanation of the top drivers.
- **Deliverables:** a notebook, a short model card, and a one-page recommendation with the expected value of the offer.
- **Data:** `telecom_churn.csv`.
- **Rubric:** a sound split and leakage check (30%); evaluation that matches the business cost (30%); explanation in plain language (25%); reproducibility (15%).

Project 3: End-to-end data engineering pipeline

- **Problem:** daily sales files arrive from several sources and must become a trusted reporting table.
- **Requirements:** raw, cleaned and reporting layers (medallion layout); idempotent loads with a key; data-quality checks (nulls, duplicates, row counts, reconciliation); a scheduled run; logs and a failure alert.
- **Deliverables:** code in a repository, a diagram of the layers, the test results, and a failure drill with notes.
- **Data:** `daily_sales.csv` split into daily files, plus a second file with customer data.
- **Rubric:** correctness and re-run safety (35%); tests and checks (25%); operations: schedule, logs, alerts (25%); documentation (15%).

Project 4: AI-powered data analyst

- **Problem:** managers ask data questions in plain English and wait days for answers.
- **Requirements:** natural-language questions turned into safe SQL against a read-only database; a row limit; the SQL shown with each answer; a test set of 30 questions with known answers; a refusal for anything outside the data.
- **Deliverables:** the working assistant, the test set and its results, and a short note on errors found.
- **Data:** `superstore_sales.csv` loaded into a database.

- **Rubric:** accuracy on the test set (35%); safety (SELECT only, limits, no personal data) (25%); transparency (SQL shown, uncertainty stated) (25%); cost per question (15%).

Project 5: Enterprise RAG knowledge assistant

- **Problem:** staff cannot find policies, procedures and past decisions across many documents.
- **Requirements:** chunking and indexing of at least 20 documents; retrieval with citations; "not found" when the answer is absent; access control so that restricted documents are never returned to unauthorised users; an evaluation of retrieval and answers.
- **Deliverables:** the assistant, an evaluation report with retrieval hit rate and citation accuracy, and an access-control test.
- **Rubric:** retrieval quality (30%); grounded answers with correct citations (30%); access control (20%); evaluation and monitoring (20%).

Project 6: AI agent for business operations

- **Problem:** a support team wants an agent that triages tickets, looks up orders and drafts replies, with approval for anything that changes a record.
- **Requirements:** read tools and at least one write tool behind approval; a step limit and budget; a regression suite of at least 30 tasks including forbidden actions; traces; cost per successful task.
- **Deliverables:** the agent, the suite and its results, the trace of a sample run, and a release checklist.
- **Rubric:** task completion on the suite (30%); safety tests passed (30%); traces and monitoring (20%); cost and latency (20%).

Project 7: MCP-powered business assistant

- **Problem:** one assistant should answer questions using a CRM, a ticket system and a help centre, through a common protocol.
- **Requirements:** an MCP server with read tools for two systems (you may use sample data), scoped permissions per tool, logging of every call, and an injection test.
- **Deliverables:** the server, a tool list with roles and fields, the logs of a demo session, and the test results.
- **Rubric:** tool design and scoping (30%); security tests (30%); logging and audit (20%); documentation for other developers (20%).

Project 8: Industry-specific capstone (choose one)

Choose an industry you know (retail, real estate, healthcare administration, logistics, education, finance operations) and combine any two earlier projects around one question in that industry. Write the problem with a named user, a decision, the data you would need and the data you can actually obtain, and deliver a working version with a clear limit statement.

- **Rubric:** relevance to the industry and the user (35%); technical quality (35%); honesty about data and limits (30%).

Portfolio checklist (every project)

- [ ] A README with the problem, the data source, how to run the project and the result.
- [ ] Code and data in a repository, with no secrets or personal data.
- [ ] Tests or checks, with the results shown.
- [ ] One chart, table or screenshot on the first page of the README.
- [ ] A clear limitations section, including what you would do next.
- [ ] A 5-minute walkthrough (video or written) that an employer can follow.
- [ ] The rubric, filled in honestly by you before you share the work.

INTERVIEW CORNER **What makes a portfolio project credible?**

A specific problem, real or realistic data, a result that is checked against something, a clear statement of limits, and evidence that you tested it. Employers can tell a project that was finished from one that was only started; the checklist is how to finish it.

::::key What to remember - Choose two projects: one for your target role and one for a different skill. - Read the rubric first and invest most effort where the marks are. - Include tests, a limitations section and a walkthrough. - A finished, honest project beats an ambitious one that is not finished. ::::

Quick quiz

5 questions · tap an answer to check it

1. How many projects should a candidate usually prepare?

A Eight, all at once

B Two: one for the target role and one for a different skill

C None

D One hundred

Answer: B. Two finished projects are more persuasive than many unfinished ones.

2. Which rubric item is common to nearly every brief?

A The colour scheme

B Correctness or quality of the core result, checked against something

C The number of pages

D The software version

Answer: B. Every brief rewards verified results.

3. Why must a RAG assistant refuse questions outside the documents?

A To save storage

B To avoid inventing answers, which is the main risk of the approach

C Because refusals are required by law

D To make the system slower

Answer: B. A grounded assistant says when the documents do not answer.

4. What belongs in the portfolio checklist's "limitations" section?

A Praise for the tool vendor

B The data gaps, assumptions and the next steps you would take

C A list of passwords

D Only the best result

Answer: B. Stating limits makes a project credible.

5. Which project is most suited to someone targeting a data engineering role?

A The AI data analyst only

B The end-to-end data engineering pipeline

C The churn dashboard only

D The MCP assistant only

Answer: B. The pipeline demonstrates reliable loading, testing and operations.

EXERCISE **Scope one project**

Choose one brief and write a one-page plan: the user and decision, the data you will use, the three deliverables, the rubric weights, and the first milestone you can finish in one week. Share the plan with someone who works in your target role and ask for one change.

Show solution (worked answer)

Hints: pick data you can actually obtain. If the brief asks for an enterprise feature, scope a smaller version and state what is missing. Put the rubric weights in your plan so that you prioritise the largest one first.

CAPSTONE, PORTFOLIO & INTERVIEWS · LESSON 9

Career Readiness Check: Know Where You Stand Before You Apply

Use a structured self-check of eight readiness statements to see whether you are ready to apply for analyst roles, which topics to close first, and how to turn each gap into a short plan. Includes the scoring bands and a plan template.

Gap-fill lesson | 2026 | Beginner | 45 min

By the end of this lesson you can

- Answer the eight readiness statements honestly, with evidence
- Interpret the score and choose the next step
- Turn each gap into a specific, time-bound action
- Set a review date and measure progress

Why a readiness check

Many people apply for jobs before they can do the work they will be asked to do, or they keep studying long after they are ready. A readiness check replaces a feeling ("I think I'm ready") with evidence ("I can write a window function from memory and I have two published projects"). It is a planning tool, not a judgement of your potential.

The eight statements

Answer **yes** only if you could demonstrate the statement today, without notes, to someone who knows the subject.

#	Statement	Evidence that proves it
1	I can write SQL joins, window functions and cohort queries from memory.	Write a cohort query on a sample database in 15 minutes
2	I can build a dashboard with a star schema and DAX measures.	A published report with at least eight measures and a data model diagram
3	I can clean data and explain every step.	A cleaning notebook or workbook, with the reasons for each change
4	I can explain a model's metrics to a non-technical person.	A two-minute explanation of MAE and R-squared for a manager, recorded
5	I have two published projects with READMEs.	Two links, each with a README that states the problem and the result
6	I can describe an ETL pipeline and its tests.	A diagram of a pipeline with the checks at each step
7	I can explain RAG and a basic agent.	A short explanation plus a working demo or a diagram of the loop
8	I can explain privacy basics for personal data.	A one-page summary of how you would handle customer data in a project

Scoring

Yes answers	What it means	Next step
10 or more of the 16 extended items (see below)	Ready to apply for analyst roles with a portfolio	Apply, keep one project in progress, and prepare for interviews
7 to 9	Close to ready	Complete the missing topics in the matching lesson and build one more project
Fewer than 7	Build the foundations first	Return to the foundation topics in the roadmap and repeat the check in a month

The eight statements above are a core check. For the 16-item version, add eight items that match your target role: for example, "I can write an Airflow DAG" for a data engineer, or "I can run a churn model with a validation set" for a data scientist. Score out of 16 and use the same bands.

::::warn Be honest with yourself A "yes" that depends on a tutorial you followed last week is not a yes. Ask: could I do this again tomorrow, without the notes? If the answer is no, mark it "not yet" and make it a task. ::::

From gap to plan

For each statement you cannot yet say yes to, write one row of a plan:

Gap	Action	Time	Evidence I will produce	Review date
Window functions	Complete the window-function exercises, then solve five new problems without notes	2 hours over two weeks	Five solved problems with explanations	Two weeks from today
Explaining metrics	Record a two-minute explanation of MAE and R-squared	1 hour	Recording reviewed by a peer	One week from today

Keep the plan to three gaps at a time. Too many gaps at once means none is closed.

Reviewing progress

Repeat the check every four to six weeks. Expect the score to move up in steps, and expect a plateau when you are learning something hard. A plateau is not failure; it means you are working on the real difficulty. Share the result with a mentor or a study partner, and ask them to check one of your "yes" answers.

Applying when you score well

When you are ready:

1. Choose three target roles and read five job descriptions for each. List the skills that repeat.
2. Match your two best projects to those skills, and put the matching skill first in your CV.
3. Prepare for the interview types in the earlier lessons: technical questions, scenarios and STAR answers.
4. Apply to a manageable number of roles each week, and track each application with its date and status.

INTERVIEW CORNER **"Are you ready for a data analyst job?"**

A strong answer names the evidence: "I have two published projects, one dashboard with a star schema and eight measures, and I can explain each measure's logic. I am working on window functions in SQL, and I am practising explanations for non-technical managers." This is honest, specific, and shows a plan.

::::key What to remember - Answer each statement with evidence, not a feeling. - Use the score bands to choose the next step: apply, close gaps, or build foundations. - Turn each gap into a time-bound action with an evidence output. - Repeat the check every four to six weeks. ::::

Quick quiz

5 questions · tap an answer to check it

1. What is the test for a "yes" answer in the readiness check?

A You have read about it

B You have watched a video about it

C You could demonstrate it today, without notes, to someone who knows the subject

D You have a certificate

Answer: C. Demonstration is the standard, not familiarity.

2. A candidate scores 8 of 16. What is the sensible next step?

A Apply to everything

B Close the missing topics and build one more project

C Stop studying

D Repeat the check the same day

Answer: B. The middle band calls for targeted gap closure and one more project.

3. Why limit the plan to three gaps at a time?

A It is required by the scoring

B Too many gaps at once mean none is closed

C Three is a legal minimum

D It makes the plan look shorter

Answer: B. Focus makes progress visible.

4. What is a good piece of evidence for "I can explain RAG and a basic agent"?

A A list of buzzwords

B A short explanation plus a demo or diagram of the loop

C A job title

D A social media post

Answer: B. Evidence shows understanding, not familiarity.

5. How often should the readiness check be repeated?

A Once a year

B Every four to six weeks

C Never, once passed

D Every day

Answer: B. Regular review shows progress and reveals plateaus.

EXERCISE **Score yourself and write the plan**

Answer the eight statements with evidence, add eight items for your target role, and score out of 16. Write a plan for your three biggest gaps, each with an action, a time, an evidence output and a review date. Share the plan with one person and ask for a challenge to one of your "yes" answers.

Show solution (worked answer)

Hints: a "yes" with no evidence attached should be marked "not yet". For the review date, put a specific calendar date in your diary. The evidence output should be something another person could check.

CAPSTONE, PORTFOLIO & INTERVIEWS · LESSON 10

A Realistic 12-Month Roadmap for a Beginner

Plan a year of learning in twelve steps, from Excel and SQL foundations through business intelligence, Python, statistics and machine learning, data engineering and cloud, GenAI and agents, to a portfolio and interview preparation. Each stage has weekly hours, a milestone, an assessment, and a plain statement of what employers generally expect.

Gap-fill lesson · ✦ 2026 · Beginner · ⏱ 90 min

By the end of this lesson you can

- Plan twelve months of learning with weekly hours and milestones
- Choose the order of topics and explain why it matters
- Set an assessment for each stage that proves the skill
- Set realistic expectations about what employers look for

Why a roadmap, and why realistic

A roadmap turns a vague wish ("I want to work with data") into a sequence of skills, each proven by a piece of work. The timing matters: a beginner who plans ten hours a week and finishes in twelve months has a real chance; a plan of thirty hours a week for three months usually stalls by the fourth week.

CASE STUDY **Two learners, one year**

Two learners start in January with the same goal. The first studies whenever motivated and skips the projects; by December she has watched many videos and has nothing to show. The second follows a roadmap of about ten hours a week, finishes a project each stage and keeps a record of each milestone. By December he has three published projects and a plan for interviews. The difference is the proof, not the hours.

The twelve months

Months	Focus	Hours per week	Milestone	Assessment
1–2	Foundation: Excel, SQL basics, basic statistics, business analytics basics	About 10	A cleaned dataset with a five-measure summary	A timed SQL test and an Excel cleaning exercise
3–4	Business intelligence: Power BI, star schema, DAX, dashboard design	About 10	A published dashboard with row-level security	Dashboard review against a rubric
5–6	Programming and data: Python, pandas, visualisation, Git and GitHub	About 12	A Python exploratory analysis of a public dataset in GitHub	Code review with tests
7–8	Machine learning and statistics: supervised learning, evaluation, a churn model	About 12	A churn model with a business recommendation	Model evaluation and explanation
9–10	Data engineering and cloud: warehouse concepts, ETL and ELT, dbt, Airflow, one cloud	About 12	A three-layer pipeline with tests	Pipeline review and a failure drill
11	GenAI, agents and MCP: LLM APIs, RAG, agents, MCP	About 12	A RAG assistant with citations and an evaluation	Evaluation report
12	Portfolio and interviews: two published projects, CV, LinkedIn, mock interviews, readiness check	About 15	Three mock interviews and two published projects	Interview rubric

The hours are a guide. A week with fewer hours is fine if the milestone is still reached; it is better to extend a stage than to skip its assessment.

Why this order

- **Excel and SQL first** because every later skill reads and writes tables. A learner who cannot join two tables cannot build a dashboard, a model or a pipeline.
- **Business intelligence before programming** because the first useful output, a dashboard, needs no code and proves the learner can answer business questions.

- **Python before machine learning** because models are built with code, and the data preparation takes most of the effort.
- **Statistics before machine learning** because the evaluation of a model is statistics; without it, a high accuracy number means little.
- **Data engineering after analytics** because pipelines exist to feed the analysis; the learner understands what the pipeline must deliver before building it.
- **GenAI and agents near the end** because they depend on the data, retrieval and evaluation skills earlier, and they are where mistakes are most expensive without those skills.

::::tip Protect the assessments An assessment is the proof that the stage is complete. A learner who skips it has consumed the content without the skill. Book the assessment in the calendar before starting each stage. ::::

What each milestone proves

Milestone	Skill it proves
Cleaned dataset with five measures	Cleaning, checking and summarising data
Published dashboard with RLS	Modelling, measures, sharing and access control
Python analysis in GitHub	Code, version control and reproducible analysis
Churn model with recommendation	Modelling, evaluation and communicating a decision
Three-layer pipeline with tests	Engineering reliability and data quality
RAG assistant with evaluation	Retrieval, grounding and measuring AI quality
Two published projects and mock interviews	Ability to show work and explain it under questioning

What employers generally expect

Employers look for working projects with code and data, clear documentation, measured results, the ability to explain trade-offs, and a steady record of learning. A roadmap cannot promise a job or a salary, and this one does not. It can make a candidate's evidence clear and specific.

WATCH OUT **Expect the first role to be a stepping stone**

Entry-level roles in analytics and data are competitive, and job titles vary between employers. A first role that uses part of the skill set is a good result, and the portfolio is what moves the next step.

Adjusting the plan

- If a stage's assessment is failed, repeat the stage's milestone before moving on.
- If your time drops, extend each stage rather than dropping the portfolio work.
- If you already know one stage, take its assessment early and move on.
- Review the roadmap at the end of each stage: what took longer, what helped, and what to change.

INTERVIEW CORNER **How would you explain your learning plan to an interviewer?**

Name the stages in order and the milestone each produced: a dashboard, a model, a pipeline, a RAG assistant. Explain why the order matters, show the two published projects, and say what you would learn next. Interviewers listen for a clear plan with evidence, not a long list of courses.

::::key What to remember - A realistic roadmap has weekly hours, a milestone and an assessment for each stage. - The order follows dependencies: data first, then analysis, programming, models, pipelines and AI. - Each milestone proves a specific skill; the portfolio makes the proof visible. - Employers look for evidence, clear documentation and explained trade-offs. - Extend a stage rather than skipping its assessment. ::::

Quick quiz

5 questions · tap an answer to check it

1. Why is business intelligence placed before programming in the roadmap?

A Power BI is easier than Python

B A dashboard proves business question skills without requiring code, and gives an early useful output

C Python is not needed

D It is the newest skill

Answer: B. The early milestone builds motivation and evidence.

2. Why should statistics come before machine learning?

A Statistics is required by law

B Evaluating a model is statistics, and a high accuracy number is meaningless without it

C Machine learning does not use data

D Statistics is shorter

Answer: B. Evaluation depends on statistical reasoning.

3. What should a learner do if an assessment is failed?

A Skip it and move on

B Repeat the stage's milestone before moving on

C Stop learning

D Change careers

Answer: B. The assessment proves the skill; repeat until it is shown.

4. Which milestone best proves interview readiness?

A A certificate

B Two published projects with mock interviews completed

C A list of videos watched

D A social media post

Answer: B. Published work and practised explanations are the evidence.

5. What does the roadmap promise?

- **A** A guaranteed job
- **B** Clear, specific evidence of skills, with no promised outcome
- **C** A salary
- **D** Certification

Answer: B. The roadmap cannot promise employment.

EXERCISE **Make your own year**

Adapt the roadmap to your situation: your weekly hours, your starting point, and the role you want. Write each stage with its months, hours, milestone and assessment. Then mark the two stages you would most likely skip under pressure and write how you will protect their assessments.

Show solution (worked answer)

Hints: start from the milestones, not the topics. If you have fewer hours, double the length of each stage rather than halving the content. Put the assessments in your calendar with dates.